

Code-Layout, Readability and Reusability

Date	6 July 2026
Team ID	SWTID-2026-7907
Project Name	EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the project	Yes	Uniform indentation followed throughout the project.
2	Proper File Structure	Files and folders are logically organize	Yes	Frontend, backend, and assets are organized into separate folders.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables are named clearly based on their purpose.
4	Function / Method Names	Functions are descriptively named	Yes	Functions use descriptive names (e.g., generateQuiz(), summarizeText()).
5	Code Comments	Inline and block comments explain log	Partial	Important functions are commented for better understanding.
6	Modular Design	Code is split into reusable functions/m	Yes	Features are separated into reusable modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate code is minimized through reusable functions.
8	Error Handling	Exceptions and errors are handled gradually.	Yes	API and user input errors are handled properly.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technolog	Where Reused	Reusability Level (High / Medium / Low)
1	Authentication Module	Python (FastAPI)	Login & Registration	High
2	Gemini API Module	Google Gemini API	Question Answering, Summarization	High
3	Quiz Generator	Python	Quiz Creation	High
4	Text Summarizer	Python	Notes & Document Summarization	High
5	Frontend UI Comonents	HTML, CSS, JavaScript	Multiple Web Pages	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-structured project folders and files.
Readability	5	Clear variable names and modular code.
Reusability	5	Reusable modules for AI features and authentication.
Documentation / Comments	4	Basic comments provided for important functions.
Overall Score	4.8 / 5	Clean, maintainable, and scalable codebase.